

FINAL INTERVENTION REPORT

The Final Intervention Report is the single source of truth for all fairness-related decisions. It documents what was done, why it was done, and how outcomes were validated.

This report represents the final stage of the fairness implementation workflow, consolidating outputs from causal analysis, intervention selection, implementation, and validation.

It provides a complete, auditable record of all fairness-related decisions before deployment.

Required Contents

The report must include:

- *Problem Description*
 - Description of observed fairness issue
 - Impacted system component
- *Affected Groups*
 - Demographic and intersectional groups impacted
- *Root Cause Analysis*
 - Summary of causal findings
 - Identified bias stage(s)
- *Intervention Strategy*
 - Selected intervention(s) (pre-, in-, post-processing)
 - Justification based on causal analysis and risk classification
 - Explanation of any skipped stages
- *Validation Results*
 - Before vs after metrics
 - Fairness improvement
 - Performance trade-offs
- *Monitoring Plan*

- Drift detection approach
- Monitoring cadence
- Ownership
- *Known Limitations*
 - Technical constraints
 - Residual risks
- *Governance Approval*
 - Risk classification reference
 - Approval authority

Referenced Documentation Artifacts

The following artifacts must be completed and attached or referenced:

- Dataset Change Log (if Pre-Processing applied)
- Model Constraint Summary (if In-Processing applied)
- Threshold Adjustment Record (if Post-Processing applied)
- Validation Summary
- Risk Classification Record

These artifacts must follow the standardized Documentation Templates.

Decision Traceability Requirement

All intervention decisions must be traceable to prior steps in the workflow:

- Causal analysis → identifies bias stage
- Intervention selection → determines mitigation strategy
- Implementation → applies selected intervention
- Validation → confirms effectiveness

Each decision must include justification and supporting evidence to ensure auditability.

Approval & Release Rules

A system may be released **only if**:

- All required stages, as determined by causal analysis and risk classification, must be completed.
- Validation passes
- Documentation is complete
- Required approvals are obtained

Final rule: If it is not documented, it is not approved.

If post-deployment monitoring detects renewed or shifted bias, the system must re-enter the intervention workflow. Deployment is invalid without completed documentation and required approvals.

Governance Enforcement

For high-risk systems:

- All decisions must be reviewed and approved by designated stakeholders
- Trade-offs between fairness and performance must be explicitly accepted
- Deployment is prohibited without formal sign-off

Governance ensures accountability beyond technical implementation.

Connection to Validation and Monitoring

This report is considered complete only after validation has confirmed:

- Measurable fairness improvement
- Acceptable performance impact
- Stability across time and groups

Post-deployment monitoring must follow the defined monitoring protocol, and any detected bias must trigger re-entry into the intervention workflow.

Certification

This report certifies that:

- The intervention is causally justified
- The intervention is proportionate to system risk
- Validation confirms fairness improvement within acceptable thresholds
- Required documentation is complete
- Required approvals have been obtained

Release authorization is granted only upon satisfaction of these conditions.